

### Question 5 (15 marks)

a) Determine if the dimensions in each of the following equations are consistent. With the variable parameters  $v$ ,  $a$ ,  $t$ ,  $x$ ,  $m$ , and  $F$  represent respectively, velocity, acceleration, time, distance, mass, and Force.

- i)  $v^2 = at$  (1 mark)
- ii)  $x = vt + \frac{1}{2}at^2$  (1 mark)
- iii)  $ma^2x + \frac{1}{2}mv^2 = Fx^2$  (1 mark)
- iv)  $v^2 = Fx/m$  (1 mark)

b) Relative to a fixed origin  $O$ , the horizontal unit vectors  $\vec{i}$  and  $\vec{j}$  are pointing due east and due north, respectively.

A particle  $P$ , of mass  $2 \text{ kg}$ , is moving under the action of two constant forces  $(2\vec{i} + 3\vec{j})\text{N}$  and  $(6\vec{i} - \vec{j})\text{N}$ .

- i) Determine, in degrees, the angle between the resultant of these two forces and the horizontal direction (4 marks)
- ii) Find the acceleration of  $P$ , in vector form. (3 marks)
- iii) If the initial velocity of  $P$  is  $(-8\vec{i} + 3\vec{j})\frac{\text{m}}{\text{s}}$ , calculate the speed of  $P$  after  $t = 3\text{s}$ .

(4 marks)

**END OF QUESTIONS**

**SECTION B: FROM THIS SECTION, CHOOSE ONLY TWO QUESTIONS (30 marks)**

**Question 3 (15 marks)**

- a) Two blocks of mass 3.50 kg and 8.00 kg are connected by a massless string that passes over a frictionless pulley (Figure 2 below). The inclines are frictionless.

Find:

- i) The magnitude of the acceleration of each block (4 marks)  
ii) The tension in the string. (3 marks)

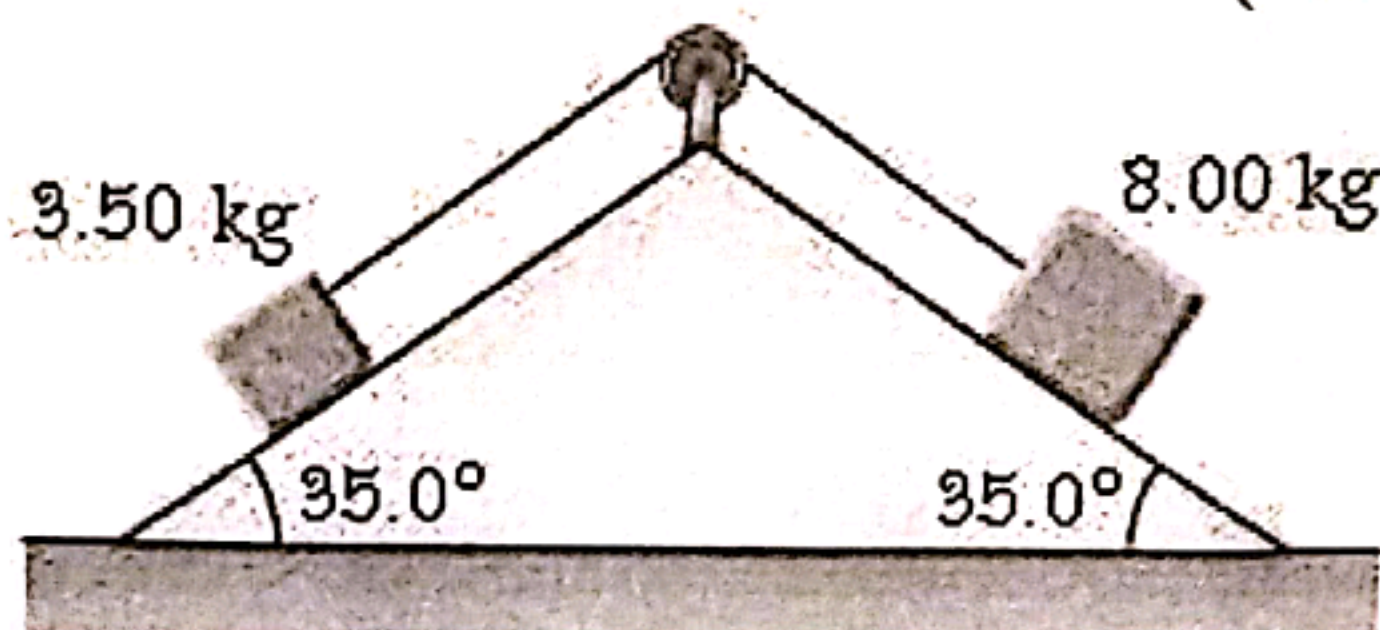


Figure 2

- b) Explain the following terms:
- i) Elastic collision and Inelastic collision (2 marks)
  - ii) Energy of a body and Power (2 marks)
- c) Explain the following physical quantities:
- i) Mass of a body (2 marks)
  - ii) Weight of a body (2 marks)

**Question 4 (15 marks)**

- a) Define
- i) An elastic body (1 mark)
  - ii) A plastic body (1 mark)
  - iii) State Work –Kinetic Energy theorem and give its mathematical expression proof. (3 marks)
- b) A small ball is projected with a speed of 20m/s at an angle of  $37^\circ$  above the horizontal from point O on the horizontal ground.
- Determine:
- i) The directional components of the initial velocity (2marks)
  - ii) The greatest height reached by the ball above the plane (2marks)
  - iii) The time of travel before it strikes the ground (2marks)
  - iv) The horizontal distance it travels in its flight (2marks)
  - v) The velocity at the maximum height (2marks)

## SECTION A: COMPULSORY QUESTIONS (20 marks)

### Question 1 (10 marks)

- a) Distinguish the scalar and vector physical quantities and give an example for each quantity. (3 marks)
- b) A particle moves along the x-axis. Its x coordinate varies with time according to the expression  $x = 2t^2 - 4t$  (m), where x is meters and t is in seconds.
- Determine the displacement of the particle in the time intervals  $t = 1\text{s}$  to  $t = 3\text{s}$  (1 mark)
  - Calculate the average velocity in the time intervals  $t = 1\text{s}$  to  $t = 3\text{s}$  (1 mark)
  - Find the instantaneous velocity of the particle at  $t = 2.5\text{s}$  (1 mark)
  - Calculate the instantaneous acceleration of the particle at  $t = 2.5\text{s}$  (1 mark)
- c) A boat moves 2.0 km east then 4.0 km north, then 3.0 km west, and finally 2.0 km south. Start at the origin and draw each vector to scale with consecutive tips from the first up to 4<sup>th</sup> tip and then find the resultant displacement. (3 marks)

### Question 2 (10 marks)

- a) i) State the Newton's laws of motion. (3 marks)  
ii) Distinguish fundamental and derived physical quantities and give an example for each Quantity. (3 marks)
- b) A traffic light weighing 100 N hangs from a vertical cable tied to two other cables that are fastened to a support. The upper cables make angles of  $37^\circ$  and  $53^\circ$  with the horizontal. Find the tension in each of the three cables as a system is in equilibrium as shown in Figure 1. (4 marks)

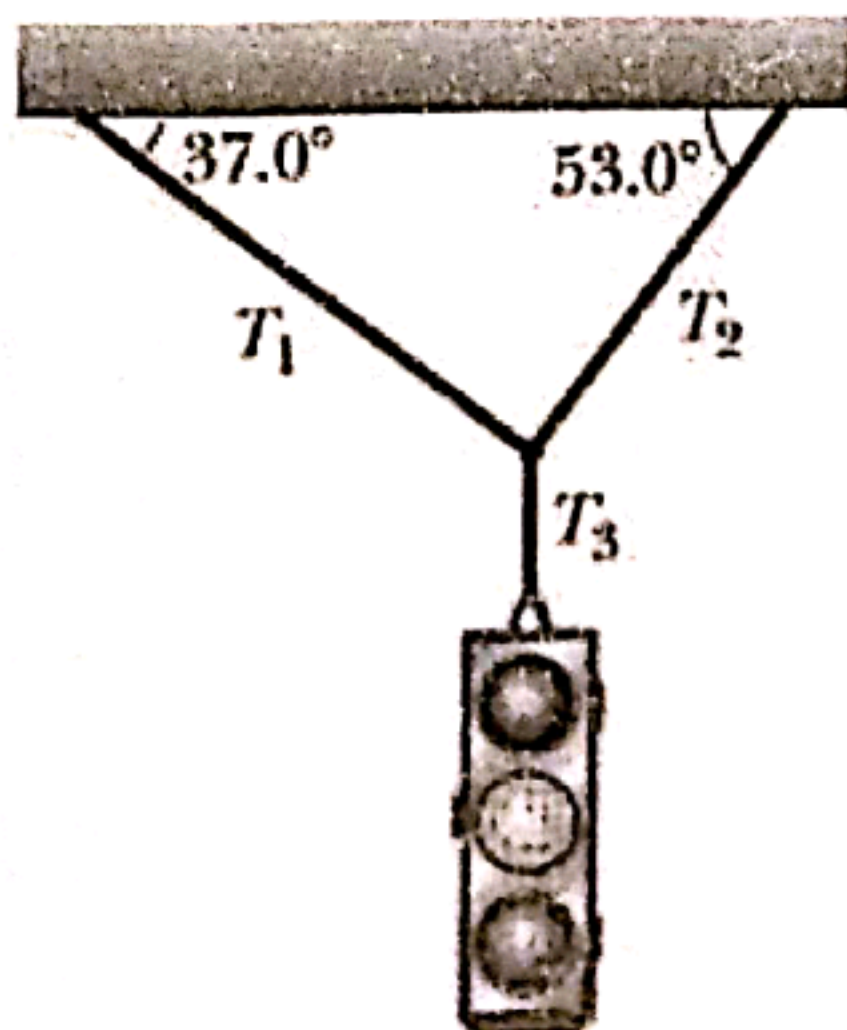


Figure 1